

Appendix

This appendix provides step-by-step proofs for the main results in the paper. Throughout, we work on the common measurable space $\mathcal{Z}^n \times \mathcal{F}$ so that both (D_n, f_T) and $(\hat{D}_n, f_S)$ are elements of the same space, and all KL divergences are well-defined whenever absolute continuity holds.

A. Preliminary Tools

A.1 Donsker–Varadhan change of measure inequality:

Lemma 1 (Donsker–Varadhan inequality). Let P, Q be probability measures on the same measurable space with $Q \ll P$. For any measurable function g with $\mathbb{E}_P[e^g] < \infty$,

$$\mathbb{E}_Q[g] \leq \text{KL}(Q\|P) + \log \mathbb{E}_P[e^g]. \quad (1)$$

Proof. Define the Radon–Nikodym derivative $r := \frac{dQ}{dP}$. Then $\mathbb{E}_P[r] = 1$ and

$$\text{KL}(Q\|P) = \mathbb{E}_Q \left[\log \frac{dQ}{dP} \right] = \mathbb{E}_P[r \log r].$$

By Jensen’s inequality applied to the convex function $\log$,

$$\log \mathbb{E}_P[e^g] = \log \mathbb{E}_P \left[r \cdot \frac{e^g}{r} \right] \geq \mathbb{E}_P \left[r \log \frac{e^g}{r} \right] = \mathbb{E}_Q[g] - \mathbb{E}_P[r \log r].$$

Rearranging gives (1). □

A.2 A standard sub-Gaussian mgf bound:

Lemma 2 (Sub-Gaussian mgf bound). Let X be σ^2 -sub-Gaussian, meaning $\log \mathbb{E}[\exp(\lambda(X - \mathbb{E}[X]))] \leq \frac{\lambda^2 \sigma^2}{2}$ for all $\lambda \in \mathbb{R}$. Then for any $\lambda \in \mathbb{R}$,

$$\log \mathbb{E}[e^{\lambda X}] \leq \lambda \mathbb{E}[X] + \frac{\lambda^2 \sigma^2}{2}. \quad (2)$$

Proof. By definition,

$$\mathbb{E}[e^{\lambda X}] = \mathbb{E} \left[e^{\lambda(X - \mathbb{E}[X])} \right] \cdot e^{\lambda \mathbb{E}[X]}.$$

Taking logs and applying the sub-Gaussian condition gives (2). □

B. Proof of Theorem 1

B.1 Statement:

Theorem 1 (Distillation generalization upper bound). Assume that $h(D_n, f_T)$ is σ^2 -sub-Gaussian under Δ_{D_n, f_T} . Then

$$\text{gen}_S \leq \text{gen}_T + \sigma \sqrt{2\mathbf{K}_n}. \quad (3)$$

B.2 Proof:

Proof. Fix any $\lambda > 0$ and choose in Lemma 1

$$P = \Delta_{D_n, f_T}, \quad Q = \Delta_{\hat{D}_n, f_S}, \quad g(d, f) = \lambda h(d, f).$$

Then (1) gives

$$\mathbb{E}_{\Delta_{\hat{D}_n, f_S}}[\lambda h(\hat{D}_n, f_S)] \leq \text{KL}(\Delta_{\hat{D}_n, f_S} \| \Delta_{D_n, f_T}) + \log \mathbb{E}_{\Delta_{D_n, f_T}}[e^{\lambda h(D_n, f_T)}].$$

By definitions, the left side equals λgen_S and the KL term equals $\mathbf{K}_n$, so

$$\lambda \text{gen}_S \leq \mathbf{K}_n + \log \mathbb{E}_{\Delta_{D_n, f_T}}[e^{\lambda h(D_n, f_T)}]. \quad (4)$$

Since $h(D_n, f_T)$ is σ^2 -sub-Gaussian with mean gen_T , Lemma 2 yields

$$\log \mathbb{E}_{\Delta_{D_n, f_T}}[e^{\lambda h(D_n, f_T)}] \leq \lambda \text{gen}_T + \frac{\lambda^2 \sigma^2}{2}. \quad (5)$$

Substitute (5) into (4):

$$\lambda \text{gen}_S \leq \mathbf{K}_n + \lambda \text{gen}_T + \frac{\lambda^2 \sigma^2}{2}.$$

Rearrange:

$$\text{gen}_S - \text{gen}_T \leq \frac{K_n}{\lambda} + \frac{\lambda\sigma^2}{2}. \quad (6)$$

Consider the function $\phi(\lambda) = \frac{K_n}{\lambda} + \frac{\lambda\sigma^2}{2}$ for $\lambda > 0$. Differentiate:

$$\phi'(\lambda) = -\frac{K_n}{\lambda^2} + \frac{\sigma^2}{2}.$$

Set $\phi'(\lambda) = 0$:

$$-\frac{K_n}{\lambda^2} + \frac{\sigma^2}{2} = 0 \implies \lambda^2 = \frac{2K_n}{\sigma^2} \implies \lambda^* = \frac{\sqrt{2K_n}}{\sigma}.$$

Plugging λ^* into (6) gives

$$\text{gen}_S - \text{gen}_T \leq \frac{K_n}{\sqrt{2K_n}/\sigma} + \frac{(\sqrt{2K_n}/\sigma)\sigma^2}{2} = \sigma\sqrt{\frac{K_n}{2}} + \sigma\sqrt{\frac{K_n}{2}} = \sigma\sqrt{2K_n}.$$

This is exactly (3). □

C. Proof of Proposition 1

C.1 Statement:

Proposition 1 (Sub-Gaussianity via stability). Assume the loss is bounded: $\ell(f, z) \in [a, b]$ for all f, z . Assume the teacher algorithm A_T is β -uniformly stable: for any neighboring datasets D_n and $D_n^{(i)}$ differing in one example,

$$\sup_{z \in \mathcal{Z}} |\ell(A_T(D_n), z) - \ell(A_T(D_n^{(i)}), z)| \leq \beta.$$

Let

$$H(D_n) := L_P(A_T(D_n)) - L_{D_n}(A_T(D_n)).$$

Then $H(D_n)$ satisfies bounded differences with constants

$$c_i = 2\beta + \frac{b-a}{n} \quad \text{for all } i \in [n], \quad (7)$$

and hence is sub-Gaussian with variance proxy

$$\sigma^2 = \frac{1}{4} \sum_{i=1}^n c_i^2 = \frac{n}{4} \left(2\beta + \frac{b-a}{n} \right)^2. \quad (8)$$

C.2 Proof:

Proof. Let $D_n = (z_1, \dots, z_n)$ and let $D_n^{(i)} = (z_1, \dots, z_{i-1}, z'_i, z_{i+1}, \dots, z_n)$ be a neighboring dataset. Define

$$f := A_T(D_n), \quad f' := A_T(D_n^{(i)}).$$

Then

$$\begin{aligned} H(D_n) &= \mathbb{E}_{Z \sim P}[\ell(f, Z)] - \frac{1}{n} \sum_{j=1}^n \ell(f, z_j), \\ H(D_n^{(i)}) &= \mathbb{E}_{Z \sim P}[\ell(f', Z)] - \frac{1}{n} \sum_{j=1}^n \ell(f', z_j^{(i)}), \end{aligned}$$

where $z_j^{(i)} = z_j$ for $j \neq i$ and $z_i^{(i)} = z'_i$.

By uniform stability and taking expectation over $Z \sim P$,

$$\left| \mathbb{E}_{Z \sim P}[\ell(f, Z)] - \mathbb{E}_{Z \sim P}[\ell(f', Z)] \right| \leq \mathbb{E}_{Z \sim P}[\ell(f, Z) - \ell(f', Z)] \leq \beta. \quad (9)$$

Consider

$$\left| \frac{1}{n} \sum_{j=1}^n \ell(f, z_j) - \frac{1}{n} \sum_{j=1}^n \ell(f', z_j^{(i)}) \right| \leq \frac{1}{n} \sum_{j=1}^n |\ell(f, z_j) - \ell(f', z_j^{(i)})|.$$

Now split the sum into $j \neq i$ and $j = i$.

For $j \neq i$, we have $z_j^{(i)} = z_j$, so by stability

$$|\ell(f, z_j) - \ell(f', z_j)| \leq \beta. \quad (10)$$

For $j = i$, we use triangle inequality:

$$|\ell(f, z_i) - \ell(f', z'_i)| \leq |\ell(f, z_i) - \ell(f', z_i)| + |\ell(f', z_i) - \ell(f', z'_i)|.$$

The first term is bounded by β by stability, and the second term is bounded by $(b - a)$ by loss boundedness. Thus

$$|\ell(f, z_i) - \ell(f', z'_i)| \leq \beta + (b - a). \quad (11)$$

Combine (10) and (11):

$$\left| \frac{1}{n} \sum_{j=1}^n \ell(f, z_j) - \frac{1}{n} \sum_{j=1}^n \ell(f', z_j^{(i)}) \right| \leq \frac{(n-1)\beta + (\beta + (b-a))}{n} = \beta + \frac{b-a}{n}.$$

So

$$|L_{D_n}(f) - L_{D_n^{(i)}}(f')| \leq \beta + \frac{b-a}{n}. \quad (12)$$

Using (9) and (12),

$$|H(D_n) - H(D_n^{(i)})| \leq \underbrace{|L_P(f) - L_P(f')|}_{\leq \beta} + \underbrace{|L_{D_n}(f) - L_{D_n^{(i)}}(f')|}_{\leq \beta + (b-a)/n} \leq 2\beta + \frac{b-a}{n}.$$

This establishes the bounded-differences constants (7).

By McDiarmid's inequality, for all $t > 0$,

$$\mathbb{P}(H(D_n) - \mathbb{E}[H(D_n)] \geq t) \leq \exp\left(-\frac{2t^2}{\sum_{i=1}^n c_i^2}\right).$$

A standard equivalence between McDiarmid tails and sub-Gaussian mgf yields that $H(D_n)$ is sub-Gaussian with variance proxy $\sigma^2 = \frac{1}{4} \sum_{i=1}^n c_i^2$. With constant c_i from (7), this gives (8). $\square$

D. Proof of Theorem 2

D.1 Central condition definition:

Definition 1 ((η, c) -central condition). A random variable X satisfies the (η, c) -central condition under P if $\eta > 0$ and $0 < c \leq 1$ and

$$\log \mathbb{E}_P[e^{-\eta X}] \leq -c\eta \mathbb{E}_P[X]. \quad (13)$$

D.2 Statement:

Theorem 2 (Distillation generalization lower bound). Assume $h(D_n, f_T)$ satisfies the (η, c) -central condition under Δ_{D_n, f_T} . Then

$$\text{gen}_S \geq c \cdot \text{gen}_T - \frac{1}{\eta} \mathsf{K}_n. \quad (14)$$

D.3 Proof:

Proof. Apply Lemma 1 with

$$P = \Delta_{D_n, f_T}, \quad Q = \Delta_{\hat{D}_n, f_S}, \quad g(d, f) = -\eta h(d, f).$$

Then

$$\mathbb{E}_{\Delta_{D_n, f_S}}[-\eta h(\hat{D}_n, f_S)] \leq \mathsf{K}_n + \log \mathbb{E}_{\Delta_{D_n, f_T}}[e^{-\eta h(D_n, f_T)}]. \quad (15)$$

By assumption, $h(D_n, f_T)$ satisfies (13) with $X = h(D_n, f_T)$ and $P = \Delta_{D_n, f_T}$:

$$\log \mathbb{E}_{\Delta_{D_n, f_T}}[e^{-\eta h(D_n, f_T)}] \leq -c\eta \mathbb{E}_{\Delta_{D_n, f_T}}[h(D_n, f_T)] = -c\eta \text{gen}_T. \quad (16)$$

Substitute (16) into (15):

$$-\eta \mathbb{E}_{\Delta_{\hat{D}_n, f_S}}[h(\hat{D}_n, f_S)] \leq \mathsf{K}_n - c\eta \text{gen}_T.$$

Divide by $-\eta$ (note $\eta > 0$ flips the inequality):

$$\mathbb{E}_{\Delta_{\hat{D}_n, f_S}}[h(\hat{D}_n, f_S)] \geq c \text{gen}_T - \frac{1}{\eta} \mathsf{K}_n.$$

Since the left side is exactly gen_S , we obtain (14). $\square$

D.4 Sub-Gaussianity implies a central condition for small η :

Remark 1 (Deriving a valid (η, c) from sub-Gaussianity). Assume X is σ^2 -sub-Gaussian with mean $\mu = \mathbb{E}[X] > 0$. Then for any $\eta > 0$,

$$\log \mathbb{E}[e^{-\eta X}] \leq -\eta\mu + \frac{\eta^2 \sigma^2}{2} = -\eta\mu \left(1 - \frac{\eta\sigma^2}{2\mu}\right).$$

Thus X satisfies the (η, c) -central condition with

$$c \leq 1 - \frac{\eta\sigma^2}{2\mu} \quad \text{provided that} \quad 0 < \eta < \frac{2\mu}{\sigma^2}.$$

E. Linear Gaussian Case Study (Detailed KL Decomposition)

E.1 Matrix normal definition and basic identities:

Definition 2 (Matrix normal distribution). A random matrix $A \in \mathbb{R}^{k \times n}$ follows a matrix normal distribution $A \sim \mathcal{MN}(M, U, V)$ if

$$\text{vec}(A) \sim \mathcal{N}(\text{vec}(M), V \otimes U),$$

where $M \in \mathbb{R}^{k \times n}$, $U \in \mathbb{R}^{k \times k}$, $V \in \mathbb{R}^{n \times n}$.

We use the vectorization identity

$$\text{vec}(WX) = (X^\top \otimes I_k) \text{vec}(W),$$

valid for $W \in \mathbb{R}^{k \times d}$ and $X \in \mathbb{R}^{d \times n}$.

For Gaussians with the same covariance, we use

$$\text{KL}(\mathcal{N}(m_1, \Sigma) \parallel \mathcal{N}(m_0, \Sigma)) = \frac{1}{2}(m_1 - m_0)^\top \Sigma^{-1}(m_1 - m_0). \quad (17)$$

E.2 Generative model: Collect features and labels column-wise into $X \in \mathbb{R}^{d \times n}$ and $Y \in \mathbb{R}^{k \times n}$ so that $D_n = (X, Y)$. Assume a noisy linear label channel

$$Y \mid X \sim \mathcal{MN}(W_* X, I_k, \nu^2 I_n),$$

where $W_* \in \mathbb{R}^{k \times d}$ is the ground-truth linear map and $\nu > 0$ is the noise level.

E.3 Teacher as a Gibbs learner (closed form): Let the teacher parameter be $W \in \mathbb{R}^{k \times d}$ with prior

$$p_0(W) = \mathcal{MN}(0, I_k, \lambda^{-1} I_d).$$

Given $D_n = (X, Y)$, define the Gibbs posterior with inverse temperature β_T :

$$q_T(W \mid D_n) \propto p_0(W) \exp\left(-\frac{\beta_T}{2\nu^2} \|Y - WX\|_F^2\right).$$

Lemma 3 (Closed form of q_T). The posterior is matrix normal:

$$q_T(W \mid D_n) = \mathcal{MN}(\bar{W}_T, I_k, \Sigma_T), \quad \Sigma_T = \left(\lambda I_d + \frac{\beta_T}{\nu^2} X X^\top\right)^{-1}, \quad \bar{W}_T = \frac{\beta_T}{\nu^2} Y X^\top \Sigma_T.$$

Proof. Let $w = \text{vec}(W)$, $y = \text{vec}(Y)$, and $B_X := X^\top \otimes I_k$. Then $\text{vec}(WX) = B_X w$ and $\|Y - WX\|_F^2 = \|y - B_X w\|_2^2$. The prior implies $w \sim \mathcal{N}(0, \lambda^{-1} I_{kd})$. Hence the (unnormalized) log density is quadratic in w with precision

$$\lambda I_{kd} + \frac{\beta_T}{\nu^2} B_X^\top B_X = \lambda I_{kd} + \frac{\beta_T}{\nu^2} (X X^\top \otimes I_k),$$

so the covariance is $(\lambda I_d + \frac{\beta_T}{\nu^2} X X^\top)^{-1} \otimes I_k = \Sigma_T \otimes I_k$. The mean is the corresponding linear term mapped back to matrix form, yielding $\bar{W}_T = \frac{\beta_T}{\nu^2} Y X^\top \Sigma_T$. $\square$

E.4 Pseudo-data generation: Sample $W_T \sim q_T(\cdot \mid D_n)$ and generate pseudo labels through the same noisy channel:

$$\hat{Y} \mid (W_T, X) \sim \mathcal{MN}(W_T X, I_k, \nu^2 I_n),$$

then define $\hat{D}_n = (X, \hat{Y})$.

E.5 Student capacity constraint via a rank bottleneck: Let the student parameter be $\Theta \in \mathbb{R}^{k \times d}$. Introduce the rank- κ map $M^*(W_T, X)$ as a best rank- κ approximation in prediction space:

$$M^*(W_T, X) \in \arg \min_{M: \text{rank}(M)=\kappa} \|W_T X - W_T M X\|_F^2.$$

Define a local Gaussian student conditional kernel

$$q_S(\Theta \mid W_T, X) = \mathcal{MN}(W_T M^*(W_T, X), I_k, \Sigma_S), \quad \Sigma_S \succ 0.$$

E.6 Process-level KL and its two terms: Define the distillation divergence

$$K_n := \text{KL}(\Delta_{\hat{D}_n, f_S} \parallel \Delta_{D_n, f_T}).$$

Using the KL chain rule on the dataset-model pair,

$$K_n = \underbrace{\text{KL}(\Delta_{\hat{D}_n} \parallel \Delta_{D_n})}_{\text{Dataset shift}} + \underbrace{\mathbb{E}_{\hat{D}_n} [\text{KL}(\Delta_{f_S|\hat{D}_n} \parallel \Delta_{f_T|D_n})]}_{\text{Algorithm shift}}. \quad (18)$$

We now bound the two terms.

E.7 Dataset shift bound and bias-variance decomposition:

a) Step 1. Condition on (X, D_n) and use convexity of KL.: Given X and D_n , $\hat{Y} \mid (X, D_n)$ is a mixture over $W_T \mid D_n$:

$$\hat{Y} \mid (X, D_n) \sim \int q_T(W_T \mid D_n) \mathcal{MN}(W_T X, I_k, \nu^2 I_n) dW_T.$$

The real label law (given X) is $\mathcal{MN}(W_* X, I_k, \nu^2 I_n)$. By convexity of KL in its first argument,

$$\text{KL}(\hat{Y} \mid X, D_n \parallel Y \mid X) \leq \mathbb{E}_{W_T \mid D_n} [\text{KL}(\mathcal{MN}(W_T X, I_k, \nu^2 I_n) \parallel \mathcal{MN}(W_* X, I_k, \nu^2 I_n))]. \quad (19)$$

b) Step 2. KL for equal-covariance matrix normals.: Vectorization gives $\text{vec}(\hat{Y}) \sim \mathcal{N}(\text{vec}(W_T X), \nu^2 I_{kn})$ and $\text{vec}(Y) \sim \mathcal{N}(\text{vec}(W_* X), \nu^2 I_{kn})$. Using (17),

$$\text{KL}(\mathcal{MN}(W_T X, I_k, \nu^2 I_n) \parallel \mathcal{MN}(W_* X, I_k, \nu^2 I_n)) = \frac{1}{2\nu^2} \|(W_T - W_*)X\|_F^2. \quad (20)$$

c) Step 3. Take expectations and split into bias and variance.: Combine (19) and (20), then average over D_n :

$$\text{KL}(\Delta_{\hat{D}_n} \parallel \Delta_{D_n}) \leq \frac{1}{2\nu^2} \mathbb{E}_{D_n} \mathbb{E}_{W_T \mid D_n} [\|(W_T - W_*)X\|_F^2]. \quad (21)$$

Write $W_T - W_* = (\bar{W}_T - W_*) + (W_T - \bar{W}_T)$ and expand:

$$\|(W_T - W_*)X\|_F^2 = \|(\bar{W}_T - W_*)X\|_F^2 + \|(W_T - \bar{W}_T)X\|_F^2 + 2\langle (\bar{W}_T - W_*)X, (W_T - \bar{W}_T)X \rangle_F.$$

Taking $\mathbb{E}_{W_T \mid D_n}$, the cross term is zero since $\mathbb{E}[W_T - \bar{W}_T \mid D_n] = 0$. Hence

$$\mathbb{E}_{W_T \mid D_n} \|(W_T - W_*)X\|_F^2 = \underbrace{\|(\bar{W}_T - W_*)X\|_F^2}_{\text{Bias}(D_n)} + \mathbb{E}_{W_T \mid D_n} \|(W_T - \bar{W}_T)X\|_F^2. \quad (22)$$

Now use the row-wise property of $W_T \mid D_n \sim \mathcal{MN}(\bar{W}_T, I_k, \Sigma_T)$. Each row has covariance Σ_T , so

$$\mathbb{E}_{W_T \mid D_n} \|(W_T - \bar{W}_T)X\|_F^2 = k \text{tr}(X^\top \Sigma_T X) =: k \text{Var}(D_n). \quad (23)$$

Plug (22)–(23) into (21):

$$\text{KL}(\Delta_{\hat{D}_n} \parallel \Delta_{D_n}) \leq \mathbb{E}_{D_n} \left[\frac{1}{2\nu^2} \left(\text{Bias}(D_n) + k \text{Var}(D_n) \right) \right]. \quad (24)$$

E.8 Algorithm shift bound and the rank-bottleneck decomposition: Define the (expected) algorithm-shift term

$$K_{\text{alg}} := \mathbb{E}_{\hat{D}_n} [\text{KL}(\Delta_{f_S|\hat{D}_n} \parallel \Delta_{f_T|D_n})].$$

Since the student kernel is conditionally Gaussian given latent W_T (and X), $q_S(\Theta \mid \hat{D}_n)$ is generally a mixture over W_T . By convexity of KL in the first argument, conditioning and then averaging yields the reduction

$$K_{\text{alg}} \leq \mathbb{E}_{D_n} \mathbb{E}_{W_T \mid D_n} [\text{KL}(q_S(\cdot \mid W_T, X) \parallel q_T(\cdot \mid D_n))].$$

d) Step 1. Closed-form KL between matrix normals.: We have

$$q_T(W \mid D_n) = \mathcal{MN}(\bar{W}_T, I_k, \Sigma_T), \quad q_S(\Theta \mid W_T, X) = \mathcal{MN}(W_T M^*, I_k, \Sigma_S).$$

Vectorize into kd -dimensional Gaussians with covariances $\Sigma_T \otimes I_k$ and $\Sigma_S \otimes I_k$. Using the standard Gaussian KL formula and Kronecker identities gives

$$\begin{aligned} \text{KL}(q_S(\cdot \mid W_T, X) \parallel q_T(\cdot \mid D_n)) &= \frac{k}{2} \left(\text{tr}(\Sigma_T^{-1} \Sigma_S) - d + \log \frac{\det \Sigma_T}{\det \Sigma_S} \right) \\ &\quad + \frac{1}{2} \text{tr}((W_T M^* - \bar{W}_T) \Sigma_T^{-1} (W_T M^* - \bar{W}_T)^\top). \end{aligned} \quad (25)$$

Define the covariance mismatch penalty

$$\text{Cov}(\Sigma_S, \Sigma_T) := \frac{k}{2} \left(\text{tr}(\Sigma_T^{-1} \Sigma_S) - d + \log \frac{\det \Sigma_T}{\det \Sigma_S} \right).$$

e) Step 2. Expand the weighted mean term using Σ_T^{-1} : From Lemma 3,

$$\Sigma_T^{-1} = \lambda I_d + \frac{\beta_T}{\nu^2} X X^\top.$$

Let $\Delta := W_T M^* - \bar{W}_T$. Then

$$\begin{aligned} \text{tr}(\Delta \Sigma_T^{-1} \Delta^\top) &= \text{tr}\left(\Delta \left(\lambda I_d + \frac{\beta_T}{\nu^2} X X^\top\right) \Delta^\top\right) \\ &= \lambda \|\Delta\|_F^2 + \frac{\beta_T}{\nu^2} \text{tr}(\Delta X X^\top \Delta^\top) = \lambda \|\Delta\|_F^2 + \frac{\beta_T}{\nu^2} \|\Delta X\|_F^2. \end{aligned} \quad (26)$$

f) Step 3. Separate rank-bottleneck residual and posterior fluctuation.: Write

$$\Delta = (W_T M^* - W_T) + (W_T - \bar{W}_T) = -W_T(I - M^*) + (W_T - \bar{W}_T),$$

and similarly

$$\Delta X = -W_T(I - M^*)X + (W_T - \bar{W}_T)X.$$

Apply $\|A + B\|_F^2 \leq 2\|A\|_F^2 + 2\|B\|_F^2$:

$$\|\Delta\|_F^2 \leq 2\|W_T(I - M^*)\|_F^2 + 2\|W_T - \bar{W}_T\|_F^2, \quad (27)$$

$$\|\Delta X\|_F^2 \leq 2\|W_T(I - M^*)X\|_F^2 + 2\|(W_T - \bar{W}_T)X\|_F^2. \quad (28)$$

Taking $\mathbb{E}_{W_T|D_n}$ and using $W_T | D_n \sim \mathcal{MN}(\bar{W}_T, I_k, \Sigma_T)$ yields

$$\mathbb{E}\|W_T - \bar{W}_T\|_F^2 = k \text{tr}(\Sigma_T), \quad \mathbb{E}\|(W_T - \bar{W}_T)X\|_F^2 = k \text{tr}(X^\top \Sigma_T X). \quad (29)$$

g) Step 4. Define the approximation and spread terms.: Define the rank-bottleneck approximation cost

$$\text{Apx}(D_n) := \lambda \mathbb{E}_{W_T|D_n} \|W_T(I - M^*)\|_F^2 + \frac{\beta_T}{\nu^2} \mathbb{E}_{W_T|D_n} \|W_T(I - M^*)X\|_F^2,$$

and the posterior spread term

$$\text{Spread}(D_n) := k \left(\lambda \text{tr}(\Sigma_T) + \frac{\beta_T}{\nu^2} \text{tr}(X^\top \Sigma_T X) \right).$$

Combining (25)–(29) with (26) and absorbing the (non-optimized) factor-2 slack from (27)–(28) into constants gives

$$\text{KL}_{\text{alg}} \leq \mathbb{E}_{D_n} \left[\text{Apx}(D_n) + \text{Cov}(\Sigma_S, \Sigma_T) + \text{Spread}(D_n) \right]. \quad (30)$$

E.9 Final compact decomposition of $\mathbf{K}_n$: Combine (18), (24), and (30):

$$\mathbf{K}_n \leq \mathbb{E}_{D_n} \left[\underbrace{\frac{1}{2\nu^2} (\text{Bias}(D_n) + k \text{Var}(D_n))}_{\text{Teacher prediction error}} + \underbrace{\text{Apx}(D_n)}_{\text{Student capacity / rank bottleneck}} + \underbrace{\text{Cov}(\Sigma_S, \Sigma_T)}_{\text{Geometry mismatch}} + \underbrace{\text{Spread}(D_n)}_{\text{Posterior spread}} \right].$$

This yields an interpretable checklist: improve teacher bias/variance to tighten dataset shift, increase effective rank κ to reduce Apx, match Σ_S to Σ_T to reduce Cov, and leverage posterior contraction to reduce Spread.

F. Proof of Theorem 3

F.1 Definitions: Let U be a random perturbation, independent of all other randomness, uniformly distributed on the Euclidean ball $\{u : \|u\| \leq \rho\}$.

For any dataset D and model f , define empirical and population sharpness:

$$\begin{aligned} S_D(f) &:= \mathbb{E}_U [L_D(f + U)] - L_D(f), \\ S_P(f) &:= \mathbb{E}_U [L_P(f + U)] - L_P(f). \end{aligned} \quad (31)$$

Define the perturbed generalization gap

$$h_U(D, f) := \mathbb{E}_U [h(D, f + U)] = \mathbb{E}_U [L_P(f + U) - L_D(f + U)]. \quad (32)$$

F.2 Statement:

Theorem 3 (Sharpness-aware distillation generalization bound). Assume:

- (i) (Local optimality in expectation) under $\Delta_{\hat{D}_n, f_S}$,

$$\mathbb{E}[L_P(f_S)] \leq \mathbb{E}[\mathbb{E}_U [L_P(f_S + U)]]. \quad (33)$$

- (ii) Under the teacher process Δ_{D_n, f_T} , both $h_U(D_n, f_T)$ and $S_{D_n}(f_T)$ are sub-Gaussian with proxies σ_u^2 and ν^2 , respectively.

Then

$$\text{gen}_S \leq \text{gen}_T + \mathbb{E}_{\Delta_{D_n, f_T}} [S_P(f_T)] + (\sigma_u + \nu) \sqrt{2\mathbf{K}_n}. \quad (34)$$

F.3 Proof:

Proof. By definition, $\text{gen}_S = \mathbb{E}_{\Delta_{\hat{D}_n, f_S}} [L_P(f_S) - L_{\hat{D}_n}(f_S)]$.

Using (33), $\text{gen}_S \leq \mathbb{E}_{\Delta_{\hat{D}_n, f_S}} [\mathbb{E}_U[L_P(f_S + U)] - L_{\hat{D}_n}(f_S)]$.

Add and subtract $\mathbb{E}_U[L_{\hat{D}_n}(f_S + U)]$ inside the expectation:

$$\begin{aligned} \text{gen}_S &\leq \mathbb{E}_{\Delta_{\hat{D}_n, f_S}} [\mathbb{E}_U(L_P(f_S + U) - L_{\hat{D}_n}(f_S + U))] \\ &\quad + \mathbb{E}_{\Delta_{\hat{D}_n, f_S}} [\mathbb{E}_U(L_{\hat{D}_n}(f_S + U)) - L_{\hat{D}_n}(f_S)]. \end{aligned}$$

Recognize the two terms using (32) and (31):

$$\text{gen}_S \leq \mathbb{E}_{\Delta_{\hat{D}_n, f_S}} [h_U(\hat{D}_n, f_S)] + \mathbb{E}_{\Delta_{\hat{D}_n, f_S}} [S_{\hat{D}_n}(f_S)]. \quad (35)$$

Let $P = \Delta_{D_n, f_T}$ and $Q = \Delta_{\hat{D}_n, f_S}$. Apply Lemma 1 with $g = \lambda h_U$ for any $\lambda > 0$:

$$\lambda \mathbb{E}_Q[h_U] \leq \mathsf{K}_n + \log \mathbb{E}_P[e^{\lambda h_U}].$$

If h_U is σ_u^2 -sub-Gaussian under P , then by Lemma 2,

$$\log \mathbb{E}_P[e^{\lambda h_U}] \leq \lambda \mathbb{E}_P[h_U] + \frac{\lambda^2 \sigma_u^2}{2}.$$

So

$$\mathbb{E}_Q[h_U] \leq \mathbb{E}_P[h_U] + \frac{\mathsf{K}_n}{\lambda} + \frac{\lambda \sigma_u^2}{2}.$$

Optimizing over λ exactly as in Theorem 1 yields

$$\mathbb{E}_Q[h_U] \leq \mathbb{E}_P[h_U] + \sigma_u \sqrt{2\mathsf{K}_n}. \quad (36)$$

Apply Lemma 1 with $g = \lambda S_D$ (the function $(D, f) \mapsto S_D(f)$) for any $\lambda > 0$:

$$\lambda \mathbb{E}_Q[S] \leq \mathsf{K}_n + \log \mathbb{E}_P[e^{\lambda S}],$$

where S denotes the random variable $S_D(f)$ under the corresponding process. If $S_{D_n}(f_T)$ is ν^2 -sub-Gaussian under P , then similarly

$$\mathbb{E}_Q[S] \leq \mathbb{E}_P[S] + \nu \sqrt{2\mathsf{K}_n}.$$

That is,

$$\mathbb{E}_{\Delta_{\hat{D}_n, f_S}} [S_{\hat{D}_n}(f_S)] \leq \mathbb{E}_{\Delta_{D_n, f_T}} [S_{D_n}(f_T)] + \nu \sqrt{2\mathsf{K}_n}. \quad (37)$$

Plug (36) and (37) into (35):

$$\text{gen}_S \leq \mathbb{E}_{\Delta_{D_n, f_T}} [h_U(D_n, f_T)] + \mathbb{E}_{\Delta_{D_n, f_T}} [S_{D_n}(f_T)] + (\sigma_u + \nu) \sqrt{2\mathsf{K}_n}. \quad (38)$$

Expand h_U using (32):

$$\mathbb{E}[h_U(D_n, f_T)] = \mathbb{E}[\mathbb{E}_U[L_P(f_T + U) - L_{D_n}(f_T + U)]] = \mathbb{E}[\mathbb{E}_U L_P(f_T + U)] - \mathbb{E}[\mathbb{E}_U L_{D_n}(f_T + U)].$$

Now add and subtract the unperturbed terms:

$$\begin{aligned} \mathbb{E}[\mathbb{E}_U L_P(f_T + U)] &= \mathbb{E}[L_P(f_T)] + \mathbb{E}[S_P(f_T)], \\ \mathbb{E}[\mathbb{E}_U L_{D_n}(f_T + U)] &= \mathbb{E}[L_{D_n}(f_T)] + \mathbb{E}[S_{D_n}(f_T)]. \end{aligned}$$

Therefore,

$$\mathbb{E}[h_U(D_n, f_T)] = \mathbb{E}[L_P(f_T) - L_{D_n}(f_T)] + \mathbb{E}[S_P(f_T)] - \mathbb{E}[S_{D_n}(f_T)] = \text{gen}_T + \mathbb{E}[S_P(f_T)] - \mathbb{E}[S_{D_n}(f_T)].$$

Substitute this into (38). The $-\mathbb{E}[S_{D_n}(f_T)]$ cancels with the $+\mathbb{E}[S_{D_n}(f_T)]$ term, leaving

$$\text{gen}_S \leq \text{gen}_T + \mathbb{E}_{\Delta_{D_n, f_T}} [S_P(f_T)] + (\sigma_u + \nu) \sqrt{2\mathsf{K}_n},$$

which is (34). $\square$

G. Derivation of the Convenient Proxies (24)–(27)

This section proves the simplified proxies stated in the main text:

$$\begin{aligned} \sigma_0 &= \frac{1}{\sqrt{n}} \left(\kappa L + \frac{b-a}{2} \right), \quad \sigma_u(\rho) = \frac{1}{\sqrt{n}} \left(\kappa(g_0 + \alpha\rho) + \frac{b-a}{2} \right), \\ \nu(\rho) &= \frac{1}{\sqrt{n}} \left(\frac{\alpha\kappa}{2} \rho + g_0 \rho + \frac{\alpha}{2} \rho^2 \right), \quad \mathbb{E}[S_P(f_T)] \leq \frac{1}{2} \tau_{\text{op}} \rho^2. \end{aligned}$$

G.1 Assumptions used: We use the following conditions, matching the main text:

- (A1) Bounded loss: $\ell(\cdot; z) \in [a, b]$.
- (A2) Parameter stability: for neighboring datasets $D, D^{(i)}$, $\|A_T(D) - A_T(D^{(i)})\| \leq \kappa/n$.
- (A3) Global Lipschitz: for all z and all f, f' , $|\ell(f; z) - \ell(f'; z)| \leq L\|f - f'\|$.
- (A4) Local regularity on $B(f_T, 2\rho)$: for all z , $\ell(\cdot; z)$ is α -smooth and $\|\nabla \ell(f_T; z)\| \leq g_0$. This implies a local Lipschitz scale $g_0 + \alpha\rho$ on $B(f_T, \rho)$.
- (A5) Population curvature: on $B(f_T, \rho)$, $\|\nabla^2 L_P(\cdot)\|_{\text{op}} \leq \tau_{\text{op}}$.

G.2 Bounded differences for the (unperturbed) teacher gap:

Lemma 4 (Bounded differences for $h(D_n, f_T)$ under (A1)–(A3)). Let $f_T = A_T(D_n)$ and $f'_T = A_T(D_n^{(i)})$ for neighboring datasets. Then

$$|h(D_n, f_T) - h(D_n^{(i)}, f'_T)| \leq \frac{2\kappa L + (b - a)}{n}. \quad (39)$$

Consequently, $h(D_n, f_T)$ is sub-Gaussian with proxy $\sigma_0 = \frac{1}{\sqrt{n}}(\kappa L + \frac{b-a}{2})$.

Proof. Write

$$h(D, f) = L_P(f) - L_D(f).$$

Then

$$|h(D_n, f_T) - h(D_n^{(i)}, f'_T)| \leq |L_P(f_T) - L_P(f'_T)| + |L_{D_n}(f_T) - L_{D_n^{(i)}}(f'_T)|.$$

By (A3),

$$|L_P(f_T) - L_P(f'_T)| = |\mathbb{E}_{Z \sim P}[\ell(f_T; Z) - \ell(f'_T; Z)]| \leq \mathbb{E}[L\|f_T - f'_T\|] = L\|f_T - f'_T\|.$$

By (A2), $\|f_T - f'_T\| \leq \kappa/n$, hence

$$|L_P(f_T) - L_P(f'_T)| \leq \frac{\kappa L}{n}. \quad (40)$$

Write

$$L_{D_n}(f_T) = \frac{1}{n} \sum_{j=1}^n \ell(f_T; z_j), \quad L_{D_n^{(i)}}(f'_T) = \frac{1}{n} \sum_{j=1}^n \ell(f'_T; z_j^{(i)}).$$

Then

$$|L_{D_n}(f_T) - L_{D_n^{(i)}}(f'_T)| \leq \frac{1}{n} \sum_{j=1}^n |\ell(f_T; z_j) - \ell(f'_T; z_j^{(i)})|.$$

For $j \neq i$, $z_j^{(i)} = z_j$, and by (A3) and (A2),

$$|\ell(f_T; z_j) - \ell(f'_T; z_j)| \leq L\|f_T - f'_T\| \leq \frac{\kappa L}{n}.$$

For $j = i$, use triangle inequality:

$$|\ell(f_T; z_i) - \ell(f'_T; z'_i)| \leq |\ell(f_T; z_i) - \ell(f'_T; z_i)| + |\ell(f'_T; z_i) - \ell(f'_T; z'_i)|.$$

The first term is at most $\kappa L/n$ by (A3)+(A2). The second term is at most $(b - a)$ by (A1). Thus

$$|\ell(f_T; z_i) - \ell(f'_T; z'_i)| \leq \frac{\kappa L}{n} + (b - a).$$

Combine:

$$|L_{D_n}(f_T) - L_{D_n^{(i)}}(f'_T)| \leq \frac{(n-1) \cdot (\kappa L/n) + (\kappa L/n + (b-a))}{n} = \frac{\kappa L}{n} + \frac{b-a}{n}.$$

So

$$|L_{D_n}(f_T) - L_{D_n^{(i)}}(f'_T)| \leq \frac{\kappa L + (b-a)}{n}. \quad (41)$$

Step 4 (finish the bounded difference). Combine (40) and (41):

$$|h(D_n, f_T) - h(D_n^{(i)}, f'_T)| \leq \frac{\kappa L}{n} + \frac{\kappa L + (b-a)}{n} = \frac{2\kappa L + (b-a)}{n}.$$

This is (39).

Step 5 (convert bounded differences into the proxy σ_0). With $c_i = (2\kappa L + (b-a))/n$ for all i , McDiarmid implies sub-Gaussian proxy $\sigma^2 = \frac{1}{4} \sum_{i=1}^n c_i^2 = \frac{n}{4} \cdot \frac{(2\kappa L + (b-a))^2}{n^2}$, hence

$$\sigma_0 = \sqrt{\sigma^2} = \frac{1}{\sqrt{n}} \left(\kappa L + \frac{b-a}{2} \right).$$

□

G.3 Bounded differences for the perturbed gap h_U :

Lemma 5 (Proxy for $\sigma_u(\rho)$ under (A1), (A2), (A4)). Under (A1), (A2), (A4), the perturbed gap $h_U(D_n, f_T)$ is sub-Gaussian with proxy

$$\sigma_u(\rho) = \frac{1}{\sqrt{n}} \left(\kappa(g_0 + \alpha\rho) + \frac{b-a}{2} \right).$$

Proof. The proof repeats Lemma 4, replacing the global Lipschitz constant L by the local Lipschitz scale $(g_0 + \alpha\rho)$ valid on the perturbation region.

Step 1 (local Lipschitz on the ball). By (A4) and smoothness, for any z and any f with $\|f - f_T\| \leq \rho$,

$$\|\nabla \ell(f; z)\| \leq \|\nabla \ell(f_T; z)\| + \alpha \|f - f_T\| \leq g_0 + \alpha\rho.$$

Thus for any such f, f' in the region,

$$|\ell(f; z) - \ell(f'; z)| \leq (g_0 + \alpha\rho) \|f - f'\|.$$

Step 2 (apply the bounded difference argument pointwise in u , then average). For each fixed u with $\|u\| \leq \rho$, apply Lemma 4 to the gap $h(D, f_T + u)$ with Lipschitz constant $(g_0 + \alpha\rho)$. This gives bounded-difference constants

$$c_i(u) = \frac{2\kappa(g_0 + \alpha\rho) + (b-a)}{n}.$$

Since this bound is uniform in u and $h_U = \mathbb{E}_U[h(D, f_T + U)]$ is an average over U , the same constants apply to h_U . Therefore h_U is sub-Gaussian with

$$\sigma_u(\rho) = \frac{1}{\sqrt{n}} \left(\kappa(g_0 + \alpha\rho) + \frac{b-a}{2} \right).$$

□

G.4 Bounded differences for empirical sharpness and the proxy $\nu(\rho)$:

Lemma 6 (Proxy for $\nu(\rho)$ under (A2), (A4)). Under (A2) and (A4), assume moreover that for any neighboring datasets $D, D^{(i)}$, if we set

$$f := A_T(D), \quad f' := A_T(D^{(i)}),$$

then the ρ -neighborhood of the line segment joining f and f' lies inside the local region on which the Hessian bound in (A4) is valid. Then the empirical sharpness

$$S_{D_n}(f_T) = \mathbb{E}_U[L_{D_n}(f_T + U)] - L_{D_n}(f_T)$$

satisfies bounded differences with constants

$$c_i^{(S)} = \frac{\alpha\kappa\rho + 2g_0\rho + \alpha\rho^2}{n}, \quad i \in [n],$$

and therefore is sub-Gaussian with proxy

$$\nu(\rho) = \frac{\sqrt{n}}{2} c_i^{(S)} = \frac{1}{\sqrt{n}} \left(\frac{\alpha\kappa}{2} \rho + g_0\rho + \frac{\alpha}{2} \rho^2 \right).$$

Proof. For each sample z , define

$$\phi_z(f) := \mathbb{E}_U[\ell(f + U; z) - \ell(f; z)].$$

Then

$$S_D(f) = \frac{1}{n} \sum_{j=1}^n \phi_{z_j}(f).$$

Let $D, D^{(i)}$ be neighboring datasets and set

$$f := A_T(D), \quad f' := A_T(D^{(i)}).$$

We bound

$$|S_D(f) - S_{D^{(i)}}(f')| \leq |S_D(f) - S_{D^{(i)}}(f)| + |S_{D^{(i)}}(f) - S_{D^{(i)}}(f')|.$$

h) Step 1: dataset replacement at fixed model.: Since only the i -th sample changes,

$$|S_D(f) - S_{D^{(i)}}(f)| = \frac{1}{n} |\phi_{z_i}(f) - \phi_{z'_i}(f)| \leq \frac{2}{n} \sup_z |\phi_z(f)|.$$

We now bound $\phi_z(f)$. For any u with $\|u\| \leq \rho$, Taylor's theorem with integral remainder gives

$$\ell(f+u; z) - \ell(f; z) = \langle \nabla_f \ell(f; z), u \rangle + \int_0^1 (1-t) u^\top \nabla_f^2 \ell(f+tu; z) u dt.$$

By (A4),

$$\|\nabla_f \ell(f; z)\| \leq g_0, \quad \sup_{\|v\| \leq 2\rho} \|\nabla_f^2 \ell(f+v; z)\|_{\text{op}} \leq \alpha,$$

hence

$$|\ell(f+u; z) - \ell(f; z)| \leq g_0 \|u\| + \frac{\alpha}{2} \|u\|^2 \leq g_0 \rho + \frac{\alpha}{2} \rho^2.$$

Taking expectation over U yields

$$|\phi_z(f)| \leq g_0 \rho + \frac{\alpha}{2} \rho^2.$$

Therefore

$$|S_D(f) - S_{D^{(i)}}(f)| \leq \frac{2g_0 \rho + \alpha \rho^2}{n}. \quad (42)$$

i) Step 2: parameter change at fixed dataset.: For each fixed z , the map $\phi_z(\cdot)$ is locally Lipschitz. Indeed, for any θ in the segment joining f and f' ,

$$\nabla \phi_z(\theta) = \mathbb{E}_U [\nabla_f \ell(\theta + U; z) - \nabla_f \ell(\theta; z)].$$

By the Hessian bound in (A4),

$$\|\nabla \phi_z(\theta)\| \leq \mathbb{E}_U [\alpha \|U\|] \leq \alpha \rho.$$

Hence ϕ_z is $(\alpha \rho)$ -Lipschitz on this segment:

$$|\phi_z(f) - \phi_z(f')| \leq \alpha \rho \|f - f'\|.$$

Averaging over the samples in $D^{(i)}$ gives

$$|S_{D^{(i)}}(f) - S_{D^{(i)}}(f')| \leq \alpha \rho \|f - f'\|.$$

Using (A2),

$$\|f - f'\| = \|A_T(D) - A_T(D^{(i)})\| \leq \frac{\kappa}{n},$$

so

$$|S_{D^{(i)}}(f) - S_{D^{(i)}}(f')| \leq \frac{\alpha \kappa \rho}{n}. \quad (43)$$

j) Step 3: combine and apply McDiarmid.: Combining (42) and (43), we obtain

$$|S_D(f) - S_{D^{(i)}}(f')| \leq \frac{\alpha \kappa \rho + 2g_0 \rho + \alpha \rho^2}{n} =: c_i^{(S)}.$$

Thus $S_{D_n}(f_T)$ satisfies bounded differences with constants $c_i^{(S)}$, and McDiarmid's inequality implies that $S_{D_n}(f_T) - \mathbb{E}[S_{D_n}(f_T)]$ is sub-Gaussian with variance proxy

$$\frac{1}{4} \sum_{i=1}^n (c_i^{(S)})^2 = \frac{n}{4} \left(\frac{\alpha \kappa \rho + 2g_0 \rho + \alpha \rho^2}{n} \right)^2.$$

Equivalently, it is sub-Gaussian with proxy

$$\nu(\rho) = \frac{\sqrt{n}}{2} c_i^{(S)} = \frac{1}{\sqrt{n}} \left(\frac{\alpha \kappa}{2} \rho + g_0 \rho + \frac{\alpha}{2} \rho^2 \right).$$

□

G.5 Bounding the population sharpness by curvature:

Lemma 7 (Population sharpness bound under (A5')). Assume

$$\sup_{\|v\| \leq \rho} \|\nabla^2 L_P(f_T + v)\|_{\text{op}} \leq \tau_{\text{op}}.$$

Then

$$\mathbb{E}[S_P(f_T)] \leq \frac{1}{2} \tau_{\text{op}} \rho^2.$$

Proof. Recall

$$S_P(f_T) = \mathbb{E}_U [L_P(f_T + U) - L_P(f_T)].$$

By Taylor's theorem with integral remainder,

$$L_P(f_T + u) - L_P(f_T) = \langle \nabla L_P(f_T), u \rangle + \int_0^1 (1-t) u^\top \nabla^2 L_P(f_T + tu) u dt.$$

Taking expectation over U and using $\mathbb{E}[U] = 0$, we get

$$\mathbb{E}[S_P(f_T)] = \mathbb{E}_U \left[\int_0^1 (1-t) U^\top \nabla^2 L_P(f_T + tU) U dt \right].$$

Since $\|U\| \leq \rho$ almost surely and $\|\nabla^2 L_P(f_T + tU)\|_{\text{op}} \leq \tau_{\text{op}}$ for all $t \in [0, 1]$,

$$U^\top \nabla^2 L_P(f_T + tU) U \leq \|\nabla^2 L_P(f_T + tU)\|_{\text{op}} \|U\|^2 \leq \tau_{\text{op}} \rho^2.$$

Therefore

$$\mathbb{E}[S_P(f_T)] \leq \int_0^1 (1-t) \tau_{\text{op}} \rho^2 dt = \frac{1}{2} \tau_{\text{op}} \rho^2.$$

□

H. Proof of Corollary 1 (Tightening Interval and Cutoff Radius ρ_0)

H.1 Baseline and sharpness-aware bounds: Define

$$B_{\text{std}} := \text{gen}_T + \sigma_0 \sqrt{2\mathcal{K}_n}, \quad B_{\text{sh}}(\rho) := \text{gen}_T + \mathbb{E}[S_P(f_T)] + (\sigma_u(\rho) + \nu(\rho)) \sqrt{2\mathcal{K}_n}.$$

We seek conditions under which

$$B_{\text{sh}}(\rho) < B_{\text{std}}.$$

H.2 Proof:

Proof. A sufficient condition for $B_{\text{sh}}(\rho) < B_{\text{std}}$ is

$$\mathbb{E}[S_P(f_T)] < (\sigma_0 - \sigma_u(\rho) - \nu(\rho)) \sqrt{2\mathcal{K}_n}.$$

By Lemma 7,

$$\mathbb{E}[S_P(f_T)] \leq \frac{1}{2} \tau_{\text{op}} \rho^2.$$

Moreover, using the standard proxy

$$\sigma_0 = \frac{1}{\sqrt{n}} \left(\kappa L + \frac{b-a}{2} \right),$$

the local proxy

$$\sigma_u(\rho) \leq \frac{1}{\sqrt{n}} \left(\kappa(g_0 + \alpha\rho) + \frac{b-a}{2} \right),$$

and Lemma 6,

$$\nu(\rho) = \frac{1}{\sqrt{n}} \left(\frac{\alpha\kappa}{2} \rho + g_0 \rho + \frac{\alpha}{2} \rho^2 \right),$$

we obtain the lower bound

$$\sigma_0 - \sigma_u(\rho) - \nu(\rho) \geq \frac{1}{\sqrt{n}} (A_0 - A_1 \rho - A_2 \rho^2),$$

where

$$A_0 := \kappa(L - g_0), \quad A_1 := \frac{3}{2} \alpha \kappa + g_0, \quad A_2 := \frac{\alpha}{2}.$$

Therefore, a sufficient condition for strict improvement is

$$\frac{1}{2}\tau_{\text{op}}\rho^2 < \frac{1}{\sqrt{n}}(A_0 - A_1\rho - A_2\rho^2)\sqrt{2\mathsf{K}_n}. \quad (44)$$

Rearranging (44) gives

$$(A_2\sqrt{2\mathsf{K}_n} + \tfrac{1}{2}\tau_{\text{op}}\sqrt{n})\rho^2 + (A_1\sqrt{2\mathsf{K}_n})\rho - (A_0\sqrt{2\mathsf{K}_n}) < 0.$$

Define

$$c_0 := A_0\sqrt{2\mathsf{K}_n}, \quad c_1 := A_1\sqrt{2\mathsf{K}_n}, \quad c_2 := A_2\sqrt{2\mathsf{K}_n} + \tfrac{1}{2}\tau_{\text{op}}\sqrt{n}.$$

Then the above condition is

$$c_2\rho^2 + c_1\rho - c_0 < 0. \quad (45)$$

If $\Delta := L - g_0 > 0$, then $A_0 = \kappa\Delta > 0$, hence $c_0 > 0$ and the quadratic in (45) has exactly one positive root,

$$\rho_0 = \frac{-c_1 + \sqrt{c_1^2 + 4c_2c_0}}{2c_2}.$$

Since $c_2 > 0$, the inequality (45) holds precisely for

$$0 < \rho < \rho_0.$$

Consequently, for every $0 < \rho < \rho_0$, the sufficient condition (44) is satisfied, and therefore

$$B_{\text{sh}}(\rho) < B_{\text{std}}.$$

□